NOT MEASUREMENT SENSITIVE

MIL-C-85322A
5 May 1989
SUPERSEDING
MIL-C-85322
9 November 1983

MILITARY SPECIFICATION

COATING, ELASTOMERIC, POLYURETHANE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for two types of coatings for exterior aircraft structures.

1.2 Classification. The coating shall be furnished in the following types and colors, as specified (see 6.2):

Type I - Rain-erosion coating

Type II - Corrosion-inhibited sealant

1.2.1 Colors. The color of the polyurethane coating shall be as listed below:

Type I	Type II
Color name	The color is to be characteristic of strontium chromate pigment. (yellow)
Gray	
Black	

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8010
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SPECIFICATIONS

FEDERAL

L-P-383	Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated
QQ-A-250/5	Aluminum Alloy Alclad 2024, Plate and Sheet,
PPP-P-1892	Paint, Varnish, Lacquer and Related Materials; Packaging, Packing and Marking of

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MIL-T-5624	Turbine Fuel, Aviation Grades JP-4 and JP-5
MIL-R-7705	Radomes, General Specification for
MIL-C-8514	Coating Compound, Metal Pretreatment, Resin-Acid
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-23377	Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-P-25421	Plastic Materials, Glass Fiber Base-Epoxy Resin, Low Pressure Laminated
MIL-R-81294	Remover, Paint, Epoxy, Polysulfide and Polyurethane Systems
MIL-T-81772	Thinner, Aircraft Coating
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft NATO Code Number H - 537
MIL-C-83286	Coating, Urethane, Aliphatic Isocyanate, for Aerospace Applications
MIL-C-85285	Coating: Polyurethane, High Solids
MIL-P-85582	Primer Coatings: Epoxy, VOC Compliant, Chemical and Solvent Resistant

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STANDARDS

FEDERAL

FED-STD-141	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing
FED-STD-313	Material Safety Data Sheets; Preparation and Submission of

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS

29 CFR 1910.1200	Material Safety Data Sheet; Preparation and Submission of
49 CFR 171-178	Regulations for the Transportation of Hazardous Materials

(Application for copies of the Code of Federal Regulations (CFR) should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Z 129.1	Precautionary Labeling of Hazardous Industrial Chemicals

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1210	Fineness of Dispersion of Pigment - Vehicle Systems
ASTM D 1640	Drying, Curing or Film Formation of Organic Coatings at Room Temperature
ASTM D 1737	Elongation of Attached Organic Coatings With Cylindrical Mandrel Apparatus
ASTM D 2196	Rheological Properties of Non-Newtonian Materials
ASTM D 2197	Adhesion of Organic Coatings
ASTM D 2370	Elongation and Tensile Strength of Free Films of Paint, Varnish, Lacquer and Related Products
ASTM D 3432	Free Toluene Diisocyanates in Urethane Prepolymers and Coating Solutions by Gas Chromatography
ASTM G 26	Operating Light and Water-Exposure Apparatus (Xenon-Arc Type) for Exposure of Nonmetallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The coatings furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3). Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under contract shall be identical, within manufacturing tolerances to the product receiving qualification.

3.2 Material. Materials specified herein shall be of such a quality as to produce products conforming to the requirements of this specification.

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3.3 Toxicity. A Material Safety Data Sheet shall be prepared and submitted in accordance with FED-STD-313 and shall meet the requirements of 29 CFR 1910.1200. One copy shall accompany the samples being submitted to the Qualifying Activity for testing (see 4.3.1). Questions pertinent to the effect of these primer coatings on the health of personnel when used for its intended purpose shall be referred by the acquiring activity to the appropriate medical service who will act as adviser to the acquiring activity (see 4.3.2 and 6.2(f)).

3.4 Composition.

3.4.1 Type I coating. The rain-erosion Type I coating shall be a two-component system consisting of Component A, a polyurethane elastomeric base component, and Component B, a ketimine type curing agent (blocked diamine). They shall be mixed in the ratio specified by the manufacturer.

3.4.2 Type II sealant. The Type II sealant shall be a two-component system consisting of Component A, a polyurethane elastomeric base component, and Component B, a ketimine type curing agent (blocked diamine). They shall be mixed in the ratio specified by the manufacturer.

3.5 Component quantitative requirements.

3.5.1 Component A. The vehicle in Component A for both Type I and II coatings shall be polyether type polyurethane prepolymers that meet the requirements of Table I. The pigmented portion shall meet the requirements in Table I when tested as specified in Table IV (see 4.6.1).

3.5.2 Component B curing agent. The curing agent, a ketimine type blocked diamine, shall be a clear, amber colored solution.

3.6 Physical properties - liquid.

3.6.1 Viscosity. The steady-state viscosity of the admixed coating shall be 50 to 500 cps when tested in accordance with Table II and 4.6.2.

3.6.2 Pot life. The pot life of the admixed coating when tested in accordance with 4.6.3, shall be at least 2 hours at 73.5 $\pm$ 2°F (23 $\pm$ 1.1°C) and 50 $\pm$ 10 percent relative humidity when maintained in a full closed container.

3.6.3 Condition in container. The coating and sealant materials shall be uniform, homogeneous and free of skins, lumps, gels or coarse particles. There shall be no separation of ingredients which cannot be dispersed readily. There shall be no gassing or swelling of the containers.

3.6.4 Storage stability. The materials shall be capable of meeting the requirements specified herein, when stored separately in their original unopened containers at a maximum temperature of 90°F (32.2°C) for a minimum of one year for Component A and six months for Component B (see 4.6).

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TABLE I. Component A requirements.

Characteristic	Type I		Type II
	Gray	Black	Yellow
Pigment (Based on Non-Volatile) (max.)	6%	6%	20%
Resin (Based on Non-Volatile) (max.)	94%	94%	80%
Fineness of Grind, PC (Mixed Topcoat) (min.)	6	6	6
Free Toluene Diisocyanate (max.) (Pigment Component)	0.4%	0.4%	0.4%
Strontium Chromate Content (min.)	--	--	9%

3.6.5 Accelerated storage stability. The pigmented components shall be subjected to accelerated storage in accordance with 4.6.4. At the end of the storage period, there shall be no gassing that causes swelling of the cans, the component materials shall be free of lumps, skins and gels and when stirred shall disperse readily to a smooth homogeneous mixture.

3.7 Physical properties - film.

3.7.1 Surface appearance. The polyurethane film shall be free from pinholes, cracks, bubbles or other film irregularities when visually examined under 10X magnification. A slight after-tack shall be permissible.

3.7.2 Drying time. The film, after application to a panel shall dry hard within a maximum time of six hours (see 4.6).

3.7.3 Free-film strength. Free-film strength of the applied coating, prepared and tested as specified in Table IV and in paragraph 4.6.5, shall meet the requirements as listed in Table II.

TABLE II. Free-film strength.

Characteristic	Requirement
Tensile Strength psi (min.)	1000
Elongation-at-break % (min.)	450

3.7.4 Adhesion.

3.7.4.1 Peel adhesion. The peel adhesion shall be a minimum of four pounds per inch of width. Film breakage at lower values than four pounds without the film peeling from the panel shall be considered passing (see 4.6.6).

3.7.4.2 Scrape adhesion. The coating shall resist removal from the primer by a 3 kg weight (see 4.6).

3.7.5 Low-temperature flexibility. The coating shall show no cracking or loss of adhesion when tested in accordance with 4.6.7.

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3.7.6 Strippability. Both newly applied, fully cured, and weathered coating shall be capable of being stripped when processed as specified in 4.6.8.

3.8 Resistance properties.

3.8.1 Water resistance. The coating after immersion in water shall show no evidence of blistering, checking or visible color change (see 4.6.9).

3.8.2 Fuel resistance. The coating after immersion in JP-4 and JP-5 fuels conforming to MIL-T-5624 shall show no evidence of blistering, checking, visible color change or other defects (see 4.6.10).

3.8.3 Fluid resistance. The polyurethane film shall withstand immersion for 4 hours at room temperature in MIL-L-23699 lubricating oil and MIL-H-83282 hydraulic fluid. Four hours after removal, the film shall not exhibit any blistering, softening, dark staining or other defects (see 4.6.11).

3.8.4 Rain-erosion resistance. Type I coating when tested as specified in 4.6.12, shall meet the minimum time erosion requirement of 30 minutes at a dry-film thickness of 12 to 14 mils (.0012 to .0014 inches).

3.8.5 Accelerated weather resistance. The coating when exposed to accelerated weathering for 500 hours, shall show no cracking, checking, embrittlement or loss of adhesion (see 4.6.13).

3.8.6 Hydrolytic stability. The film shall show no softening or loss of adhesion when tested as specified in 4.6.14.

3.8.7 Electrical transmission. The coating shall meet the requirements as specified in Table III (see 4.6.15).

TABLE III. Electrical transmission.

Characteristic	After 7 day cure at 50 ± 10% RH and 75 ± 5°F	After 7 day cure plus 7 days at 90% RH and 100°F
Electrical Transmission (% min.)	90	85

3.9 Working properties.

3.9.1 Application. Type I and II coatings shall be suitable for brush or spray application to plastic laminates, composites and aluminum alloy substrates. The material shall be mixed in accordance with the manufacturer's instructions. It shall achieve a minimum dry-film thickness of 2 mils per pass.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.2.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the test method document or the applicable paragraph of this specification.

4.3 Qualification inspection. Qualification inspection shall consist of all the tests specified in Section 3.

4.3.1 Qualification samples. The test samples shall consist of at least three quarts of the coating Type I material in the gray and black colors and/or the yellow colored Type II material. The material shall be furnished in containers of the type to be used in filling contract orders. Samples shall be identified as follows and forwarded to the laboratory designated in the letter of authorization (see 6.3).

Qualification test samples
Specification MIL-C-85322A (Type I, Gray or Black
or Type II, Yellow) (as applicable)
Coating, Elastomeric, Polyurethane
Manufacturer's name and product number
Submitted by (name and date) for qualification testing in
accordance with authorization (reference authorizing letter)

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4.3.2 Test report. In addition to the qualification test samples, the manufacturer shall furnish a test report showing that the material satisfactorily conforms to the requirements of this specification. The manufacturer shall furnish to the contracting activity the toxicological data and formulations required to evaluate the safety of the material for the proposed use through the submission of the Material Safety Data Sheet detailed in FED-STD-313 and 29 CFR 1910.1200.

4.3.3 Retention of qualification. In order to retain qualification of a product approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that the manufacturer's product complies with the requirements of this specification. The time of periodic verification by certification shall be in two-year intervals from the date of original qualification. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.4 Quality conformance inspection.

4.4.1 Lot information. A lot shall consist of all the coating of the same color, manufactured at one-time from one batch, forming part of one contract, and submitted for acceptance. A batch shall consist of all coating material manufactured during one continuous operation and forming part of one contract or order for delivery.

4.4.2 Physical property testing. The quality conformance inspection shall consist of Component A resin solids (3.5.1), fineness of grind (3.5.1) and viscosity (3.6.1) determinations prior to delivery. The Government reserves the right to conduct tests for storage stability and weather resistance or to conduct any or all tests of this specification at any time within six months from the date of manufacture of the coating as attested by the date appearing on the container's label. Samples for test shall be taken from previously unopened containers.

4.4.3 Examination of packaging, packing and marking. An examination shall be made to determine that packaging, packing and marking comply with the requirements of Section 5 of this specification. Defects shall be scored in accordance with the list below. Shipping containers fully prepared for delivery that have not been palletized shall be examined for defects of closure. See 6.9 for suggested sampling and inspection levels.

Examine	Defect
Packaging	Container not as specified, closures not accomplished by specified or required methods or materials.
	Leakage or seepage of contents. Non-conforming component, component missing, damaged or otherwise defective. Bulged or distorted container.
Markings	Data, including directions for use, omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

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4.4.4 Examination for palletization. An examination shall be made to determine that palletization complies with the requirements of Section 5 of this specification. Defects shall be scored in accordance with the list below. See 6.10 for suggested sampling and inspection levels.

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.

4.5 Test panels. Test panels for all tests except; peel adhesion, electrical transmission and rain-erosion shall be aluminum clad aluminum alloy conforming to QQ-A-250/5, anodized in accordance with MIL-A-8625, Type I. Panels shall be 3 by 6 by 0.020 inches (76 by 152.4 by .51 mm) in size. The primer for glass fabric base plastic laminates shall be MIL-C-8514, wash primer. The primer for aluminum substrates shall be MIL-P-23377 or MIL-P-85582, Type I epoxy primer.

4.5.1 Application of primer and coating. The film thickness of the wash primer shall be 0.3 to 0.5 mils and the epoxy primer shall be 0.6 to 0.9 mils. The primer shall dry for 60 minutes before application of the coating. The dry-film thickness of the Type I coating shall be 12 to 14 mils and the Type II sealant shall be 4 to 6 mils. The coating shall be applied at a rate of two mils per pass. Cure time shall be 7 days before testing.

4.6 Test methods. The tests of this specification shall be conducted in accordance with Table IV and paragraphs 4.6.1 through 4.6.15 and the panels used shall be prepared as specified in 4.5. Ingredient material submitted shall be tested to determine compliance with the applicable specification. Test conditions shall be as specified in the applicable test method or paragraph. Unless otherwise specified herein, all tests shall be conducted at a temperature of 73 ± 5°F and a relative humidity of 50 ± 10 percent.

TABLE IV. Test methods.

Requirements paragraph	Test	FED-STD-141 Test Method No.	ASTM Method No.
3.5.1	Free Toluene Diisocyanate of Prepolymer		D 3432
3.5.1	Resin solids	4051	
3.5.1	Fineness of grind		D 1210
3.5.1	Total Pigment	4021	

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TABLE IV. Test methods (continued).

Requirements paragraph	Test	FED-STD-141 Test Method No.	ASTM Method No.
3.6.1	Viscosity		D 2196
3.6.4	Storage stability	3022	
3.7.2	Drying time		D 1640
3.7.3	Free-film strength		D 2370
3.7.4.2	Adhesion scrape		D 2197 1/
3.7.5	Low-temperature flexibility		D 1737
3.8.5	Accelerated weathering		G 26

1/ Method A

4.6.1 Pigment.

4.6.1.1 Strontium analysis. In order to determine that the chromate inhibitor contained in the pigmented portion of Component A is strontium chromate, the dried sample from the total pigment determination (4.6.1.3) shall be analyzed with an emission spectrograph in the "arc" condition. The spectra obtained shall be compared with that of a pigment blend known to meet Table I in a qualitative manner. The sample shall exhibit a line density equivalent to or better than the control at wavelengths at 2931.83°A and 3380.71°A.

4.6.1.2 Chromate, total.

4.6.1.2.1 Reagents.

a. Sulfuric Acid, H_2SO_4, concentrated (98%).

b. Hydrogen Peroxide, H_2O_2, 30%.

c. Deionized water.

d. Fischer SO-C-192 1000 ppm Cr AA standard or equivalent.

4.6.1.2.2 Procedure.

a. For this procedure a Perkin Elmer 403 atomic adsorption unit or equivalent, equipped with a nitrous oxide-acetylene or an air-acetylene head, shall be used. Set up the atomic absorption unit for the standard operation parameters given for chromium. (wavelength: 357.9, slit setting: 0.7 nm, current to hollow cathode lamp: 28 ma, flame type: nitrous oxide-acetylene for PE 403 AA)

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b. Standards for chromium are prepared by suitable dilutions of the stock standard (Fischer SO-C-192 1000 ppm Cr or equivalent). Aliquots of the standards, as well as a reagent blank, should be treated similarly to the samples. The standard range is .5 ppm to 10 ppm Cr for this analysis.

c. Film samples should be cut into small pieces approximately 1/3 inch square. Liquid samples should be weighed directly into the 125 ml Phillips beaker. Place an accurately weighed 0.1 to 0.2 gram portion of sample into a dry 125 ml phillips beaker and add approximately 4 ml of H_2SO_4 with the 10 ml graduate. Cover with a watchglass. Place the beaker on a medium temperature hotplate until the sample chars and white fumes of sulfuric acid are evolved. Remove the beaker from the heat and allow the sample to cool for two minutes. Remove the watchglass cover and very cautiously add 7ml of H_2O_2 dropwise, at a rate of 30-40 drops per minute. Recover the beaker with the watchglass and place the sample back on the hotplate. Continue heating until the sample is completely digested and all of the H_2O_2 is evolved as noted by the disappearance of the yellowish-brown color. After cooling, transfer to a 100 ml volumetric flask and dilute to mark with deionized water.

d. Directly aspirate the solutions and determine their concentration using the standard conditions (from (a)). Make suitable dilutions of the samples, if necessary to bring the sample concentrations into appropriate range (from (b)).

e. Calculate as follows:

$$\text{ppm Cr in sample} = \frac{(\text{conc. of sample in ppm})(\text{volume in ml})(\text{d.f.})}{\text{sample weight in g}}$$

$$\text{where d.f.} = \frac{\text{final dilution volume in ml}}{\text{ml of aliquot taken for dilution}}$$

4.6.1.3 Total pigment. The percentage of pigment in Component A is found using Method 4021 of FED-STD-141 with the following exceptions:

a. The extraction mixture shall be acetone (A.C.S. reagent grade).

b. The solids shall be dried in a vacuum oven at 105 $\pm$ 2°C (221 $\pm$ 3.6°F) and no more than 10 mm Hg for two hours.

4.6.2 Viscosity. Allow the admixed coating to stand undisturbed for 30 minutes. The viscosity shall be tested in accordance with ASTM D 2196 using a Brookfield viscosimeter (Model LVT, Spindel #2, 30 RPM) or equivalent. Readings should be made when a steady-state is achieved.

4.6.3 Pot life. The Type I and the Type II components shall be mixed as per manufacturer's instructions. The coatings shall be stored in a full closed container for 2 hours at room temperature. The working properties and surface appearance of a panel sprayed with the material shall meet all the requirements of Section 3, except for storage stability (3.6.4) and accelerated storage stability (3.6.5) requirements.

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4.6.4 Accelerated storage stability. Unopened containers of the pigmented components shall be subjected to a minimum of 4 days of accelerated storage at 120 ± 5°F. At the end of the 4 day storage period, the cans shall be visually inspected for swelling and gassing. The cans shall be opened and inspected for lumps, skins and gels; and the ease of the dispersing by paddle to a smooth homogeneous mixture shall be observed (see 3.6.5).

4.6.5 Free-film strength. Two 8-inch by 10-inch sheets of TEDLAR or vinyl film shall be coated with the test coating as specified in 4.5.1. After 4 days cure, the coated films shall be placed in lukewarm water and the free-film of coating separated. After blotting dry with paper towels, the films shall be cured and tested in accordance with ASTM D 2370 for the requirements of 3.7.3.

4.6.6 Peel adhesion.

4.6.6.1 Peel adhesion test panels. Test panels for the peel adhesion test shall be flat, glass-fiber base laminates conforming to any type and fabric number of MIL-P-25421. The glass laminate panels shall be of low void content, dense and representative of high quality plastic laminated structures. They shall be lightly sanded with 180-240 grit abrasive paper to remove the glossy finish and wiped clean with a clean cloth saturated in a solvent such as toluene or methyl ethyl ketone to remove dust and contamination before applying any primer or coating material.

4.6.6.2 Peel adhesion method. Three inch by six inch by approximately 0.04 inch fiberglass panels shall be used for this test. After sanding and solvent cleaning as specified in 4.6.6.1, a three by three inch part of the panel shall be covered with TEDLAR film approximately 0.002 inch thick held down at the exposed fiberglass surface by a 1/2 inch piece of pressure sensitive cellulose tape on both sides. The entire surface shall be coated as specified in 4.5.1. After 4 days, the Tedlar film and cellulose pressure-sensitive tape shall be carefully removed. Two panels shall be cured for a total of 168 hours. The panels shall be immersed in water for 4 hours. Four parallel razor blade cuts through to the fiberglass shall be scribed in the panel crossing the nonadhered and adhered coating sections and placed at 1/4 inch, 1-1/4 inch, 1-3/4 inch and 2-3/4 inch from the edge of the panel. One 1 inch free strip of coating shall be folded back over the adhered coatings and the uncoated panel and free strip of coating shall be placed in the jaws of a Scott Tensile Tester or equivalent to provide 180° pull at a speed of 2 inches per minute.

4.6.7 Low-temperature flexibility. Two panels as specified in 4.5 shall be coated and room-temperature cured as specified in 4.5.1. The coated panels shall then be conditioned for a minimum of 1 hour at -60°F and while at that temperature bent 180 degrees over a 1/4 inch mandrel in accordance with ASTM D 1737. The panels shall be examined immediately after bending for evidence of failure, then conditioned at 70 ± 5°F, and reexamined for conformance to 3.7.5.

4.6.8 Strippability. The coatings shall be applied to flat, glass-fiber base laminates conforming to MIL-P-25421 and cured as specified in paragraph 4.5.1. Apply MIL-R-81294, Type IV chemical stripper and allow to stand approximately 15 minutes or until blistering and lifting occurs. Remove the softened coating using a plastic scraper. Reapply stripping material to

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remove any residual primer or topcoat. Rub the test panel surface with a clean cloth saturated with methyl ethyl ketone (MEK) to clean the coating residue from the panel. The panel shall meet the requirements of paragraph 3.7.6.

4.6.9 Water resistance. Test panels as specified in 4.5 shall be coated as specified in 4.5.1. After 2 to 10 hours air-dry, they shall be topcoated with 2.0 ± 0.3 mils of MIL-C-83286 or MIL-C-85285 polyurethane coating and cured for seven days. The panels shall then be immersed in distilled water for 4 days at 120 ± 5°F. At the end of the immersion period, the specimens shall be removed from the water and examined for conformance to 3.8.1.

4.6.10 Fuel resistance. Four coated panels shall be prepared and cured, as specified in 4.5.1. The panels shall be immersed in JP4 and JP5 fuel conforming to MIL-T-5624 for a period of 4 hours at room temperature. At the end of the period, the specimens shall be removed from the fluids and visually examined for conformance to paragraph 3.8.2.

4.6.11 Fluid resistance test. Six coated panels shall be prepared and cured, as specified in 4.5.1. The panels shall be immersed in each of the following for 4 hours: MIL-L-23699 lubricating oil and MIL-H-83282 hydraulic fluid. Four hours after removal, the films shall be visually examined for conformance to paragraph 3.8.3.

4.6.12 Rain-erosion resistance. Four round disks (1 inch in diameter by 3/16 inch thickness) cut from an epoxy-fiberglass laminate conforming to MIL-P-25421 shall be used for testing each color (gray, black and white). The coating shall be applied to the test panels as specified in paragraph 4.5.1 and cured as specified. Each coated disk shall be mounted in a horizontal whirling-arm tester at the end of the rotating arm, so that the disk is perpendicular to the direction of travel. A suitable water ring, mounted above the rotating blade, shall be used to simulate a natural rainfall of one-half inch per hour and two millimeter droplet size. The test shall be run at 500 miles per hour at the sample's center (800 kilometers/hrs). Determine the average time in which the 4 specimens of the same color and type have eroded through to the substrate (see 3.8.5).

4.6.13 Accelerated weather resistance. Test panels prepared as directed in 4.5 and 4.5.1, shall be exposed for 500 hours in a 6000 watt Xenon-arc weatherometer (Atlas Electric Devices Company or equivalent) that is cycling between 102 minutes of light only and 18 minutes of light and waterspray. Coatings shall be examined to determine conformance to the requirements of 3.8.5. The following conditions shall apply when tested according to ASTM G 26, Type BH:

Black body temperature in cabinet: 140 ± 5°F
Percent relative humidity in cabinet: 50 ± 5
Intensity of Xenon-arc: 0.3-0.4 watts/sq. meter at 340 nm wavelength

4.6.14 Hydrolytic stability. Two panels, as specified in 4.5, shall be coated and room temperature cured as specified in 4.5.1. The panels shall be stored at 200 ± 5°F and 95 percent relative humidity for 4 days. A glass desiccator jar may be used for storage of the panels. The panels shall be placed, coated side up, on a tray in the desiccator. The desiccator shall

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contain a glycerine (22 percent by weight) in water solution in the bottom, which will produce a relative humidity (RH) of 95 percent at the test temperature. The desiccator shall then be closed and inserted into an air circulating oven maintained at 200 ± 5°F. At the end of the exposure period inspect the panels visually and check for softening of the film by pushing a thumbnail or spatula over the film (see 3.8.6).

4.6.15 Electrical transmission.

4.6.15.1 Test panels. Test panels shall be approximately 24 inches by 34 inches by 0.165 ± 0.01 inch thick. They shall be fabricated using 16 layers of No. 181E fiberglass and polyester resin conforming to L-P-383. The fiberglass to polyester ratio shall be 62 to 38 parts by weight. The laminate shall be dense, of low void content and representative of high quality laminate fabrication. The dielectric constant of the laminate shall be 4.20 ± 0.05. Coating shall be applied to the previously sanded and solvent wiped panel in accordance with 4.5.1 and 4.6.6.1. One panel shall be used as a control panel and shall be uncoated.

4.6.15.2 Sequence of testing. The sequence of testing the panels shall be as follows:

a. Coated panels shall be cured in accordance with 4.6.15.1.

b. The uncoated control panel and one coated panel shall be tested as specified below.

c. The same panels specified in (b) above shall be conditioned at a temperature of 100 ± 5°F and a relative humidity of 90 ± 5 percent for 7 days and then tested as specified below.

4.6.15.3 Transmission test equipment. Test equipment performance requirements for flat panel samples shall meet the transmission efficiency test of MIL-R-7705.

4.6.15.4 Test procedure. The test shall be conducted at a frequency of 9.375 (GHz) gigahertz. The test sample shall be clamped at the approximate midpoint between the horns at a 60° incident angle. The horns shall be adjusted so that polarization is horizontal. The incident angle shall be varied slightly on either side of 60° to maximize power transmission. Minimum and maximum power transmission shall be determined through a 3/8 inch lateral movement of the test panel and averaged. The average of the two values shall be considered the power transmission.

4.6.15.5 Test data and computation. The test data shall be converted to and reported, as percent transmission, using the equation listed below (see 3.8.7):

$$\frac{T2 \text{ sample}}{T2 \text{ blank}} \times 100 = \text{percent transmission}$$

Where T2 sample = power transmission of coated panel

T2 blank = power transmission of uncoated panel

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5. PACKAGING

5.1 Packaging and packing. The coatings shall be packaged, packed and marked in accordance with PPP-P-1892. The level of packaging shall be A or C and the level of packing shall be A, B or C as specified (see 6.2). The size of the containers shall be either 1 quart, 1 gallon or five gallons. The coating shall be supplied in a kit packaged as a unit, consisting of two or more cans of the same colored "Component A" (resin portion) and one can of "Component B" (curing agent).

5.2 Marking and labeling. In addition to the marking specified in PPP-P-1892, individual cans and containers shall bear a printed label showing the following nomenclature and information as applicable:

Component Identification
Component A - Pigmented polyurethane resin component
Component B - Curing agent component
Specification MIL-C-85322A Type I (Gray or Black) or II (as applicable)
Manufacturer's name and product number
Date of manufacture by month and year
Batch number/Net contents

All unit and intermediate packs of toxic and hazardous chemicals and materials shall also be labeled in accordance with the applicable laws, statutes, regulations or ordnances, including Federal, state and municipal requirements. In addition, unit and intermediate containers, including unit containers that serve as shipping containers such as pails and drums, shall be marked with the applicable precautionary information detailed in American National Standard ANSI Z 129.1.

5.2.1 Precautionary markings. Unit, intermediate and shipping containers shall contain the following precautionary marking: "WARNING! Contains flammable volatile solvent." Shipping containers shall be marked in accordance with Department of Transportation regulations as specified in 49 CFR 171-178 and shall bear the "Flammable Liquid" red label as specified therein.

Precautions: (To be included on a sheet with each kit)

a. Open pigmented component carefully. Some outgassing may be evident. Discard if can is bulged.

b. Clean spray equipment and hoses thoroughly, before and immediately after use, with methyl ethyl ketone.

c. Mix only that amount which can be used within 2 hours in accordance with manufacturer's instructions.

d. Wear protective clothing, creams, gloves and goggles while spraying.

e. An air-line respirator is recommended for spraying operations. If it is not available, protective masks containing an organic-vapor cartridge respirator should be used.

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f. Do not contaminate Components A or B with alcohol or water.

g. Keep containers closed when not in use. Both components are moisture sensitive.

h. Wash hands immediately after use.

5.2.1.2 Lid (Component A only). Each lid of Component A material shall bear a red printed label with the following information:

"USE CAUTION
OPEN SLOWLY"

6. NOTES

6.1 Intended use. The Type I material covered by this specification is intended as a rain-erosion coating for aircraft radomes and leading edges. The Type II material is utilized as a flexible sealant to protect fastener patterns on the exterior surfaces of aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Classification (Type I or II).

c. Type and size of kit desired, including quantity and size of containers (see 5.1).

d. Color number and name. Requisitions, contracts and orders should specify the quantity and size of container or containers in which the polyurethane coating is to be furnished.

e. Level of packaging and packing (see 5.1).

f. Special markings (see 5.2)

g. Toxicological data requirements (see 3.3 and 4.3.2).

h. FAR clauses 23.303 and 52.223-3.

i. Specify if palletization is required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this

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specification. The activity responsible for the Qualified Products List is the Naval Air Development Center, Attn: Code 6062, Warminster, PA 18974; and information pertaining to qualification of products may be obtained from that activity. It is understood that the material furnished under this specification subsequent to final approval shall be of the same composition and shall be equal to products upon which approval was originally granted. In the event that the coating furnished under contract is found to deviate from the composition of the approved product or that the product fails to perform satisfactorily, approval of such products will be subject to immediate withdrawal from the Qualified Products List.

6.4 Subject term (keyword) listing.

Chromate inhibitor
Coating
Corrosion inhibitive
Elastomeric
Exterior
Flammable
Flexible
Hazardous material
Material Safety Data Sheets
Metal surfaces
Polyurethane
Qualification
Qualified Products List (QPL)
Rain-erosion coating
Sealant
Spray techniques

6.5 Moisture. The polyurethane components should be kept dry. The presence of moisture degrades the quality of the paint. Packaging of the materials should be done in a dry atmosphere. Solvents for the resins should be examined for evidence of contamination before they are incorporated even though they are of urethane grade. Urethane grade solvents or thinners may become contaminated with water in tank cars or storage tanks. The purchase of urethane grade solvents or thinners is no guarantee that excessive moisture is not present. It is therefore recommended that all users check for moisture contamination. The following suggested method may be used to determine the presence of water: add one drop of aluminum secondary butoxide to 100 ml of the solvent in a stoppered flask and shake. An appreciable amount of turbidity indicates the presence of water. The pigments used in the topcoats must be absolutely dry before being incorporated into the resin solution. If not, the package stability will suffer.

6.6 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313 and meeting the requirements of 29 CFR 1910.1200. The pertinent government mailing addresses for submission of data are listed in Appendix B of FED-STD-313.

6.7 Chromium content by direct aspiration flame atomic absorption interferences. In the air-acetylene flame, the absorption of chromium is suppressed by iron and nickel. If analysis is performed in a lean flame, the

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interferences will be lessened but the sensitivity will be reduced. The nitrous oxide-acetylene flame eliminates most chemical and/or matrix interferences that occur in the air-acetylene flame with only a slight reduction in sensitivity.

6.8 Chromium content by direct aspiration flame atomic absorption note. The standardization range of .5ppm to 10ppm Cr is intended for analysis with the nitrous oxide-acetylene flame. Use of the air-acetylene flame would lower the linear range and possibly necessitate greater sample dilutions.

6.9 Examination of packaging, packing and marking sampling and inspection levels. The sample unit for this examination shall be one shipping container fully prepared for delivery except that it shall not be palletized and need not be sealed. The lot size shall be the number of shipping containers in the end item inspection lot. The samples for this examination shall be selected at random in accordance with MIL-STD-105, inspection level S-2 and acceptable quality level (AQL) 4.0 defects per hundred units.

6.10 Examination for palletization sampling and inspection levels. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The samples for this examination shall be selected at random in accordance with MIL-STD-105, inspection level S-1 and acceptable quality level (AQL) 6.5 defects per hundred units.

6.11 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - AS
Air Force - 20

Review activities:
Army - AV, ME

User activities:
Navy - MC

Preparing Activity:
Navy - AS
(Project No. 8010-1226)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85322A	COATING, ELASTOMERIC, POLYURETHANE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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